

Chapter 5.

$$5.1. (a) PV = (110 - 100)(e^{-0.06})^2 - 10 \\ = -1.1308$$

$$(b) PV = 0 - 10 \\ = -10$$

$$5.2. (a) PV = 0 - 5 = -5$$

$$(b) PV = (100 - 98)(1 + \frac{0.06}{12})^{-6} - 5 \\ = -3.059$$

Method 1:
5.3. By the put-call parity formula.

$$S + P - C = Ke^{-rt} \quad (t=1)$$

Since $K < \min S_t$, we know that

$$P = 0. \text{ Then } C = S - Ke^{-r}$$

Method 2:

Since $K < \min S_t$, the call option will be simply exercised. One investment is to purchase a call option at time 0 and exercise it at time 1. The present value of the cost is $C + Ke^{-r}$. Another investment is to purchase it directly at time 0. The present value of the cost is S . Both investment results in owning the security at time 1. By the law of one price, we have (in order to avoid arbitrage)

$$S = C + Ke^{-r}$$

$$C = S - Ke^{-r}$$

5.4. If $C > S$, we have an arbitrage by simultaneously buying the security and selling the call option.

5.5 We know

$$S + P - C = Ke^{-rt}$$

$$\Rightarrow C = S + P - Ke^{-rt}$$

$$\geq S - Ke^{-rt} \quad (\text{since } P \geq 0)$$

Thus

$$C \geq 30 - 28e^{-0.05/3} = \underline{2.4624}$$

5.7. (a) Not necessarily true.

For example, when $K \geq S$, we may have $P \geq S$.

(b) True.

$$S + P - C = Ke^{-rt}$$

We know from P5.4 that $C \leq S$. Thus

$$P = Ke^{-rt} + (C - S)$$

$$\leq Ke^{-rt}$$

$$\text{Since } |e^{-rt}| \leq 1$$

$$\Rightarrow P \leq K$$

(Another explanation is that if $P > K$ we have an arbitrage by simultaneously selling the put option)

5.8. By the put-call parity formula

$$P = Ke^{-rt} - S + C$$

$$\geq Ke^{-rt} - S \quad (\text{since } C \geq 0)$$

5.10. Proof:

Investment 1: Buying the security, buying the put option, and selling the call option at time 0.

$$\text{Initial cost: } S + P - C$$

$$\text{yields at time } t: \begin{cases} K & \text{if } S(t) \leq K; \\ & (\text{exercise the put}) \\ K & \text{if } S(t) > K \\ & (\text{the call will be exercised}) \end{cases}$$

Investment 2: put Ke^{-rt} in the bank at time 0.

Initial cost: Ke^{-rt}

yields at time t : $(Ke^{-rt})e^{rt} = K$

By the Law of one price, We have

$$S + P - C = Ke^{-rt}$$

otherwise, there is an arbitrage.

5.11 (a) K (you will exercise the put option for sure since $K > S_1 > S_2$)

(b) Buy the put-call parity formula

$$S + P - C = Ke^{-rt}$$

Since $K > S_1 > S_2$, we know the call option is worthless, i.e., $C = 0$

thus

$$S + P = Ke^{-rt}$$

$$P = Ke^{-rt} - S$$

5.12: We set up two investment:

Investment 1: Buying the call option and the put option at time 0.

Initial cost: $C_1 + C_2$

yields at time t : 1.

Investment 2: Put e^{-rt} in the bank at time 0

Initial cost: e^{-rt}

yields at time t : 1.

Thus by the Law of one price, either there is an arbitrage or

$$C_1 + C_2 = e^{-rt}$$

5.13.

$$S + P - C = 25 + 3 - 3 = 25$$

$$Ke^{-rt} = 20e^{-0.1/4} = 20 \cdot e^{-0.025} = 19.506$$

$$\Rightarrow S + P - C > Ke^{-rt}$$

Thus an arbitrage is obtained by selling the security, selling the put, buying the call.

Initial yield: 25

Cost after three month: 20.

The present value of gain is $25 - 20e^{-0.025} = 5.494$

5.14 let C and P be the costs of the European call and put options. We know

$$\begin{cases} C_a = C, P_a \geq P \\ S + P - C = Ke^{-rt} \end{cases} \Rightarrow$$

$$S + P_a - C_a \geq Ke^{-rt}$$

5.15. First, we must have $P_1 > P_2$ since if $P_1 < P_2$ an arbitrage is effected by buying the P_1 -put and selling the P_2 -put.

Furthermore, if $K_1 - K_2 < P_1 - P_2$ then an arbitrage is effected by selling the P_1 -put and buying the P_2 -put;

$$\begin{cases} \text{Initial yield: } P_1 - P_2 \\ \text{Cost at time } t: K_1 - K_2 \text{ or } 0. \end{cases}$$

$\Rightarrow$ A sure win is obtained since $P_1 - P_2 > K_1 - K_2 > 0$

Thus we must have $K_1 - K_2 \geq P_1 - P_2$

5.16 let $t > S \geq 0$. let P_t be the price of the Put option with expiration time t , and P_S for time S .

If $P_t \leq P_S$, one could buy the expiration time t put and sell the expiration time S put.

If the sold put is ever exercised, then you should exercise the bought put at the same time, which cancel each other, so you have the initial difference in prices as your arbitrage.

5.20. (a) $(S(t) - K)^+ + (K - S(t))^+ = |S(t) - K|$

(b) $(S(t) - K_1)^+ - (K_2 - S(t))^+$

(c) $2(S(t) - K_1)^+ - S(t)$

(d) $S(t) - (S(t) - K)^+$